A Survey on Deep Learning for Monte Carlo Path Tracing

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Recent strides in hardware-accelerated ray tracing have propelled algorithms once deemed suitable only for offline rendering, like Monte Carlo path tracing, into interactive frame rates. While path tracing has been regarded as a practical utility in animating scenes for the film industry, achieving visually noise-free imagery often mandates thousands of samples per pixel and considerable computation time. Regrettably, this poses a difficulty for video games and virtual reality applications, which demand high frame rates and resolutions, thereby constraining the computational overhead of path tracing. Two extant approaches, in-process sampling, and post-processing reconstruction methods, i.e., denoising and upsampling, address this challenge. The giant evolution of deep learning technology has emerged as pivotal in path tracing processing. We explore and advance Monte Carlo path tracing technology based on deep learning. Moreover, we illustrate the merits and demerits of diverse designs and technologies, propose potential future development trends, and aim at providing researchers with a comprehensive understanding of the cutting-edge in deep learning-driven Monte Carlo path tracing.

CCS Concepts: • **Computing methodologies → Machine learning**; **Ray tracing;**

Additional Key Words and Phrases: Monte Carlo, path tracing, deep learning, sampling techniques, denoising, upsampling

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1 Introduction

Simulating the physical behavior of light in computer graphics has long been a challenge. **Monte Carlo** (**MC**) path tracing approaches provide an approximate solution. Renowned for its precision and adaptability, it has emerged as the primary algorithm for physics-based rendering across the animation and visual effects sector [92]. The advent of hardware-accelerated ray tracing GPUs [2, 5, 129] can produce noisy low-sample-count renderings at interactive rates, also accelerated using linear algebra hardware. Yet, when tackling movie rendering or demanding scene requirements, attaining noise-free images remains time-intensive, spanning from hours to days. Consequently, the ongoing refinement of physics-based rendering algorithms is a pivotal research endeavor.

Deep learning has emerged as an up-and-coming technology, finding applications in various challenging tasks such as image classification [95], pattern recognition [171], natural language processing [33], and healthcare [15]. Its integration into MC path tracing has significantly improved rendering effects compared to traditional approaches. With the continuous advancement of deep learning technology, revolutionary changes have been witnessed in the MC path tracing field, resulting in substantial improvements in rendering quality and speed.

1.1 An Overview of Monte Carlo Path Tracing

MC path tracing employs random sampling to simulate the propagation and reflection of light within three-dimensional scenes. By tracing the path of rays through the scene, the algorithm calculates the color value for each pixel in the resulting image. From the perspective of the camera, the algorithm gradually calculates the intersection points between rays and objects in the scene. Considering the material properties and lighting conditions at each intersection point, determine the color value for that specific point. Through recursive tracing of light paths, the algorithm ultimately determines the color value for each pixel, generating realistic images based on physics principles. The rendering equation [86] Equation (1), illustrated in Figure 1, governs light transport on surfaces:

$$L_0(x, \vec{\omega_0}) = L_e(x, \vec{\omega_0}) + \int_{\Omega_{2\pi}} f_r(x, \vec{\omega_0}, \vec{\omega_i}) \cos(\theta_i) L_i(x, \vec{\omega_i})\, d\omega_i. \tag{1}$$

L_e represents the radiance emitted directly by a surface, while f_r denotes the material's **bidirectional reflectance distribution function** (**BRDF**) or **bidirectional scattering distribution function** (**BSDF**). $\Omega_{2\pi}$ integrates over the hemispherical domain. The cosine term $\cos(\theta_i)$ accounts for the projected area of the incident radiance $L_i(x, \vec{\omega_i})$ onto the surface normal. In path tracing, a ray iteratively interacts with scene surfaces until intersecting a light source or meeting a termination condition. The final pixel value is computed by recursively accumulating radiance contributions, weighted by the BRDF f_r and viewing direction $\vec{\omega_i}$.

For a one-dimensional interaction $\int_a^b f(x)\mathrm{d}x$, MC estimation uses uniform random variables $X_i \in [a, b]$. The estimator is

$$F_N = \frac{b-a}{N} \sum_{i=1}^{N} f(X_i). \tag{2}$$

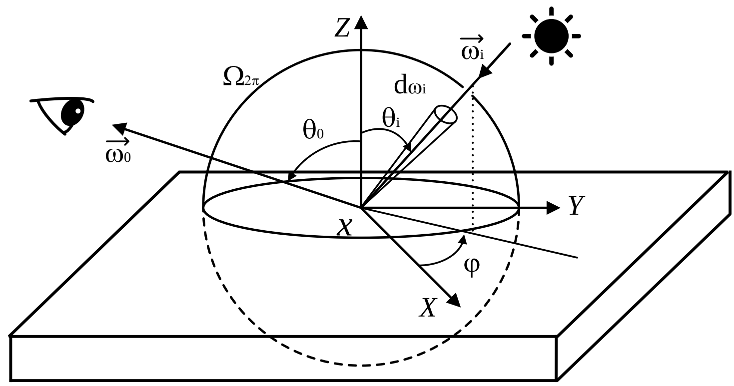

Fig. 1. Schematic of the rendering equation.

$E[F_N]$, is in fact to the interaction. The **probability density function** (**PDF**) $p(x)$ associated with the random variable X_i must be equal to 1/(b-a) since p must both be a constant and also integrate to one over the domain $[a, b]$.

If the random variable X_i are drawn from any PDF $p(x)$, the Equation (3) can be used to estimate the interaction:

$$F_N = \frac{1}{N}\sum_{i=1}^{N}\frac{f(X_i)}{p(X_i)}. \tag{3}$$

For a light path x = $x_0 \ldots x_l$ in the path space P (with x_0 on an emitter and x_l on a sensor), the pixel intensity I_p is expressed as

$$I_p = \int_p h(x)f(x)\,dx, \tag{4}$$

where f(x) is the measurement contribution function and h(x) is the pixel reconstruction filter. The MC approximation with N path samples x_k is

$$\langle I_p \rangle = \frac{1}{N}\sum_{k=1}^{N}\frac{h(x_k)f(x_k)}{p(x_k)}. \tag{5}$$

The estimator is unbiased if: (i) Complete coverage: p(x) $>$ 0 whenever f(x) $\neq$ 0; (ii) Weight correction: Contributions are scaled by $\frac{f(x_i)}{p(x_i)}$; (iii) Consistency: $\lim_{n\to\infty}\langle I_p\rangle = I_p$.

$$MSE(\hat{\theta}) = E(\hat{\theta} - \theta)^2. \tag{6}$$

By the property of unbiasedness, $MSE(\hat{\theta})$ equals the estimator's variance $Var(\hat{\theta})$. Theoretical analysis demonstrates that $MSE(\hat{\theta})$ converges to zero at a rate of $\mathcal{O}(1/N)$ as the sample size N increases. Correspondingly, the estimator's standard deviation $\sqrt{Var(\hat{\theta})}$ converges at $\mathcal{O}(1/\sqrt{N})$. Consequently, minimizing error for a fixed N remains a critical research topic in physics-based rendering despite its extensive history.

Path tracing has become a cornerstone of physics-based rendering in industry, remaining a 40-year-old algorithm that continues to inspire research. However, alternative algorithms like **bidirectional path tracing** (**BDPT**) have yet to see widespread adoption. This trend stems from three key factors: (i) Algorithm Efficiency: PT samples light paths from the camera via MC integration, offering simplicity and speed in direct lighting scenarios. BDPT improves complex indirect lighting and caustics but incurs higher computational costs for direct-light-dominated scenes. (ii) Implementation Complexity: PT uses straightforward recursive/iterative ray tracing, while BDPT requires managing dual sub-paths (camera and light) and dynamic connection strategies, increasing memory and design complexity. (iii) Industry Support: PT is integrated into standard tools like NVIDIA OptiX 7 [128], whereas BDPT lacks unified frameworks, forcing developers to build custom solutions. Despite the potential for future hardware acceleration to enhance BDPT, conventional PT remains the most cost-effective solution today. Nevertheless, integrating BDPT with advanced techniques like ReSTIR [66] is gaining traction for real-time path tracing applications.

Since many MC samples are required for a high quality image, this leaves four main approaches for making rendering faster: (i) making samples faster to evaluate (e.g., GPU rendering, ray tracing hardware, optimized low-level algorithms), (ii) being clever in choosing the light paths to sample (i.e., sampling techniques), (iii) denoising or reconstruction, attempting to produce a better picture out of the samples by relying on various smoothness assumptions or analytic models of the transport phenomena being modeled [94], and, finally, (iv) improving the resolution of MC rendered images by upsampling. Our survey primarily concentrates on the MC path tracing based on deep learning. Still, it does not address content related to (i), as it pertains to hardware research, which is outside the scope of our discussion. Furthermore, this survey also incorporates recent advancements in rendering based on deep learning research.

1.2 Previous Surveys

Several articles reviewed MC path tracing, ray tracing, and deep learning-related work. Deng et al. [31] reviewed hardware accelerators and microarchitecture technologies for real-time ray tracing. Meister et al. [113] comprehensively surveyed the BVH acceleration structures in modern ray tracing, focusing on construction algorithms and different hardware accelerations. Hua and Gruson et al. [71] oriented gradient-domain light transport to accelerate MC rendering, leveraging a gradient-based estimation and reformulating the rendering problem as an image reconstruction. Zwicker et al. [186] reviewed adaptive sampling and reconstruction algorithms in MC rendering based on non-machine learning approaches. Huo et al. [77] reviewed relevant research on MC denoising based on deep learning. Zhu [182] conducted extensive and detailed investigations into the fundamental principles of machine learning and its applications, such as denoising, path guiding, rendering participating media, and other difficult light transport situations. Kang et al. [89] introduced radiation estimation, photo relaxation, and so on, giving readers an overall introduction to photo mapping and motivating further research. Tewari et al. [150] offered a comprehensive insight into fundamental computer graphics and machine learning principles, delving into pivotal facets of neural rendering. Wang et al. [157] delved into deep learning-enhanced rendering methods, surveying pertinent studies integrated into the traditional rendering pipelines with neural networks, i.e., the forward subset of the neural rendering. These reviews have summarized and organized relevant research, offering valuable references for researchers.

Given the recent developments, an updated review of current research becomes essential for researchers to stay abreast of the latest advancements in the field. Many literature reviews and research studies on deep learning and MC path tracing already exist, providing knowledge from previous years. A comprehensive review by Huo et al. [77] has inspired our survey. In contrast to singularly emphasizing denoising methods, our survey offers a holistic exploration of deep learning techniques with MC path tracing, which includes insights into sampling techniques based on deep learning alongside prevalent upsampling methods found in contemporary gaming and development trends. We aim at advancing the development of MC path tracing based on deep learning, ultimately striving to achieve real-time, high-quality path tracing on user platforms.

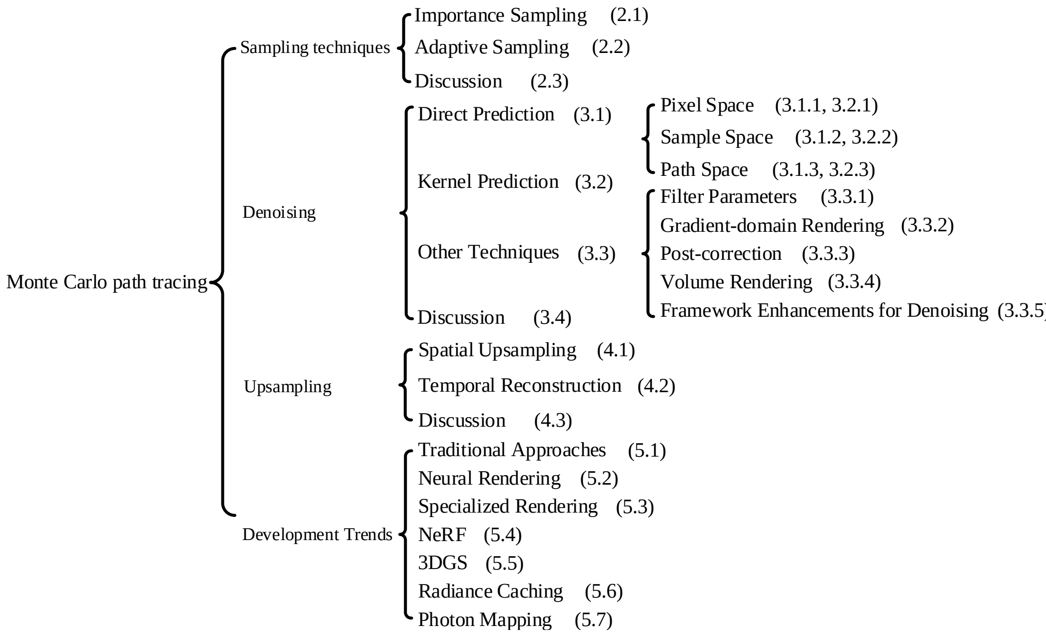

Fig. 2. Organization of this survey.

1.3 Article Organization

Figure 2 shows the organization of this survey. The remainder of this survey is structured as follows. Section 2 introduces a classification of sampling techniques. Each technique category is designated as subsections (2.1, 2.2). Section 3 delves into MC rendering denoising techniques based on deep learning, categorizing them primarily into direct prediction and kernel prediction (Sections 3.1, 3.2) methods. Based on their data requirements, these are subdivided into pixel, sample, and path space. This section synthesizes key findings from various studies, including in-depth analyses of this significant research. It also discusses other techniques (Section 3.3). In Section 4, upsampling techniques are discussed and categorized into spatial and temporal domains (Sections 4.1, 4.2). We discussed sampling techniques, denoising, and upsampling methods separately (Sections 2.3, 3.4, 4.3). Section 5 outlines recent development trends. Lastly, this survey concludes with a summary in Section 6.

2 Sampling Techniques

MC path tracing can only sample a finite number of random optical paths to compute an image. Consequently, all MC methods are susceptible to variance in the estimated pixel values, which manifests as image noise. However, achieving a noise-free image often requires computation times extending to several hours. In response, researchers have developed various sampling techniques over the years, with deep learning-based methods, including **importance sampling** and **adaptive sampling**, becoming particularly prominent for improving image quality.

2.1 Importance Sampling

As shown in Equation (5), the estimator's variance is not only inversely proportional to N, but also has a strong dependence on the shape of $p(\mathrm{x})$. When $p(\mathrm{x})$ closely matches the normalized $h(\mathrm{x})\mathrm{f}(\mathrm{x})$, the variance is minimized; conversely, significant disparities between the two lead to higher variance. Importance sampling optimizes the estimation by aligning $p(\mathrm{x})$ with $h(\mathrm{x})\mathrm{f}(\mathrm{x})$, enabling efficient and precise probability density estimation for a given sample X_i.

Kajiya [86] discussed the concept of importance sampling to reduce variance in MC rendering. Over the past few decades, researchers have explored two primary types of importance sampling techniques. The first type was incremental sampling, which involved local importance sampling in each step during the incremental construction of the optical path. Veach and Guibas [153] introduced multiple importance sampling to combine the advantages of individual path sampling techniques. Clarberg [28] utilized wavelet representation to perform importance sampling of products of environmental maps, BRDFs, and visibility products. The second technique was global sampling. Metropolis light transport [154] treats the global path space as a single sample in a global path space. The method also defines the target density. Various path space parameterizations–such as Markov chains methods [99]–have been proposed to enhance the sampling process further [133].

Dahm et al. [30] introduced a consistent algorithm that used reinforcement learning [142] to progressively learn where light comes from. This approach substantially increased the number of non-zero contribution light transport paths while shortening the average path length and enhancing overall efficiency. Zheng et al. [179][1] proposed an importance sampling technique employing neural networks to learn how to sample from a desired density defined by a set of samples. Their approach treated an existing MC rendering algorithm as a black box. They leveraged the neural network architecture specifically tailored to represent **real-valued non-volume preserving** (**"Real NVP"**) [36] transformations in high-dimensional spaces. Bako et al. [10] developed the first offline, machine learning approach for importance sampling in path space, called ODIS. This method effectively processed the first-bounce light paths for general scenes, enabling path sampling with as little as 1 spp rate. They demonstrated the first practical use of **generative adversarial networks** (**GANs**) [49] within the MC rendering pipeline. Nevertheless, its efficacy was limited as it only supports guiding the first bounce, which proved ineffective in scenes dominated by indirect lighting.

Müller et al. [121] created using deep neural networks for sample generation in MC integration, leveraging **non-linear independent component estimation** (**NICE**) [35]. They derived a gradient-descent-based optimization framework for the KL and the χ^2 divergence for the specific application of MC integration with unnormalized stochastic estimates of the target distribution. Their methodology enables the light-transport simulation learning of path-sampling densities and importance sampling of multi-dimensional path prefixes within the **primary sample space** (**PSS**), i.e., the space of random numbers used to generate paths. They demonstrated learning of joint path-sampling densities in the PSS and the importance sampling of multi-dimensional path prefixes thereof. However, in path guiding applications, the primary cost of their method arose from the evaluation of coupling layers, which resulted in excessive time over-head, particularly in simple scenes. Müller et al. [122] built upon their previous work and introduced an unbiased variance reduction parameter for MC interaction with **neural control variables** (**NCV**). They employed a normalizing flow to approximate the shape of the integrand equation alongside a second neural network designed to infer the solution of the integration equation. This approach effectively found a high-quality approximation of the integrand with minimal integration cost.

Specific components of the rendering equation–such as the BRDF or direct lighting–can be importance-sampled. However, when indirect lighting dominates, sampling efficiency tends to degrade. Since 2021, path guiding based on deep learning has emerged as a pivotal approach to address this challenge, becoming a key area of focus in importance sampling. A path guiding method usually learns distributions over the 3D scenes that better fits the rendering equation. Then, the path tracer can sample the scattering direction, improving rendering efficiency more effectively.

[1] https://quan-zheng.github.io/publication/impsamplepss19/

Zhu et al. [184] introduced a neural path guiding approach that harnessing photons traced from light sources as the primary input for sampling density reconstruction. This method utilized offline trained neural networks to reconstruct high-fidelity path guiding sampling distributions from a sparse set of samples. They partitioned the scene space into an adaptive hierarchical grid, with the application network reconstructing the high-quality sampling distributions for any local region in the scene. Zhu and Xu et al. [183] adopted a deep neural network tailored to reconstruct accurate hierarchical high-resolution sampling distributions efficiently. They leveraged an offline-trained neural network to accelerate online learning in traditional path guiding, employing the quad-tree based representation for efficient high-resolution distribution modeling. This approach facilitated more fine-grained directional sampling, effectively enhancing the practicality and efficiency of neural path guiding. Dong et al. [37] presented **Neural Parametric Mixtures** (**NPM**), a neural formulation for encoding target distribution in path guiding algorithms. NPM utilized a continuous and compact implicit representation to encode spatial-directional target distributions and decode them into **Parameter Mixing Models** (**PMM**) with lightweight neural networks for fast importance sampling. Their research effectively guided the modeling of the target distribution (incident radiation or product integrand) for path guiding.

Huang et al. [74] proposed an online framework that utilized a compact neural network and stochastic ray samples to model the spatially varying distribution of the full light transport product in the rendering equation. The learned distribution enabled efficient sampling of the full incident light transport product. They introduced the **Normalized Anisotropic Spherical Gaussian Mixture** (**NASG**) model, which was trained online via a small MLP-based neural network [149] and auxiliary ray samples, ensuring both efficiency and effectiveness in capturing complex lighting behaviors. Lu et al. [105] proposed **Voxel Path Guiding** (**VXPG**), a real-time path guiding method that significantly improves fitting efficiency even with limited samples. The key innovation lies in reusing a shared spatial irradiance representation across shadow points, storing irradiance and geometry data in boundary-aware voxels. Each frame begins by building the voxel structure and sampling shading points, selecting a voxel and mapping the shadow point to its interior, and grouping voxels into supervoxels using a streamlined K-means [104] algorithm based on visibility and irradiance.

2.2 Adaptive Sampling

Adaptive sampling involves estimating a sampling map through neural networks using low-sample MC rendering inputs. This map is then used to strategically allocate additional samples, enhancing the sampling process's targeting. Data-driven neural approaches to adaptive sampling have been shown to outperform traditional non-neural counterparts by a significant margin, offering more efficient and accurate results.

Early approaches to adaptive sampling within the image plane relied on metrics such as contrast [117], confidence bounds of the mean [148], or image discontinuities [12]. While straightforward to implement, these methods could not adjust for variance or bias introduced by denoising techniques. Consequently, they often resulted in sub-optimal sampling maps when aiming at minimizing rendering error after denoising. Over the past decade, significant efforts have been invested in exploring the balance of adaptive sampling and denoising. While numerous methods have been proposed, many adaptive sampling methods are specifically tailored to the denoising method with which they are coupled [18, 100, 120, 137, 139].

There are exceptions where arbitrary reconstruction is feasible. For instance, one adaptive sampling is based on empirical two-buffer variance after denoising [137, 139]. However, this approach necessitates multiple independent images and individual denoising, limiting its practicality due to potential noise variance estimation issues and an inability to detect denoising biases stemming from inadequate sampling. Another prevalent adaptive sampling method relies on SURE (Stein's Unbiased Risk Estimate) [144]. Unfortunately, SURE requires calculating the partial derivative of the denoiser's output with its input during inference, incurring considerable computational costs. Alternatively, some adaptive sampling methods operate on additional dimensions other than the image plane [59] or exploit the analytical properties of the integrand function to determine sampling budgets [39]. Deep learning has achieved remarkable success in denoising MC renderings in recent years, spurring research into adaptive sampling grounded in deep learning methodologies.

Dachsbacher et al. [29] were innovative in integrating machine learning into rendering, introducing methodologies for classifying various visibility configurations and showing their application of existing visibility-based approaches. Their approach utilized co-occurrence matrices for visibility analysis, extending this analysis to clusters on the surfaces of triangles rather than individual pixels. This method facilitated perception-driven detail-level techniques in real-time rendering applications by detecting configurations with expected visual masking. Kuznetsov et al. [97][2] proposed DASR, the first learning-based adaptive sampling approach for MC rendering, delivering better results in general scenes featuring soft shadows, depth of field, and global illumination compared to traditional sampling techniques. Their methodology involved modeling the estimation process of the sampling map using a CNN. This CNN took the 1 spp noisy images along with several auxiliary features as the input, subsequently utilizing the sampling map to allocate an additional 3 spp.

Vogels et al. [155] referenced [100, 120, 137] and proposed an approach utilizing two iterations. The first iteration started with 16 spp (non-adaptive). Subsequently, each iteration involved denoising the image and executing an error-predicting network that predicted error maps. The next iterations doubled the total sample count across all pixels, allocating them to pixels proportionally to the predicted error maps. Huo et al. [78] designed a method for training quality and reconstruction networks (Q- and R- networks, respectively) using an extensive set of offline datasets for adaptive sampling and reconstruction of the initial reflected incident radiation field. The **deep Q-network** (**DQN**) [119], leveraging deep reinforcement learning, predicted and guided the adaptive sampling process. Hasselgren et al. [62] expanded upon the end-to-end training method proposed by [97] to incorporate the temporal domain, significantly enhanced both the efficiency and temporal stability of the sampling technique. Their study introduced temporal optimization to the sample predictor, enabling it to learn spatio-temporal sampling strategies. The adaptive sampling network utilized the warped previous denoised output (warped using application-provided motion vectors) along with feature buffers for the current frame (normals, depth, motion vectors, and albedo at first hit).

Salehi et al. [140] proposed an innovative adaptive sampling and reconstruction technique for offline MC rendering. Their approach involved sampling through the analysis of noise distributions, effectively acquiring training data by leveraging minimal per-pixel statistical data from the target distribution. In terms of adaptive sampling, their methodology shared similarities with DASR, yet they approximated the distribution of average values using a readily sampled analysis distribution, enabling the direct generation of sample averages. By synthesizing new renderings from the analysis noise distribution, they sidestepped the need to store and load sample count cascades, resulting in accelerated training and reduced data dependency. Firmino et al. [43] presented an adaptive sampling technique, requiring no additional neural networks or learning beyond previous methods. They proposed a novel denoising output variance estimation based on first-order Taylor series expansion of the denoiser and computing forward mode automatic differentiation and the Jacobian-vector product. This estimation variance was then utilized to compute the sample distribution for the subsequent rendering iteration.

[2] http://cseweb.ucsd.edu/~viscomp/projects/dasr/

Table 1. Methods for Sampling Techniques

	Ref.	Year	Archit.	Description
Importance Sampling	[30]	2017	RL	Introduce an algorithm to improve light transport simulation.
	[179]	2018	CNN	Perform a non-linear warp in the PSS.
	[10]	2019	GAN	Train a model to reconstruct full incident radiance from sparse samples.
	[121]	2019	CNN	Propose piecewise-polynomial transforms and the one-blob encoding.
	[122]	2020	RL	Propose NCV for unbiased variance reduction in MC integration.
	[184]	2021	CNN/ED	Leverage photons traced from light source as the primary input.
	[183]	2021	CNN/ED	Combined path samples with photons as hybrid input data.
	[37]	2023	MLP	Encode distributions via a compact implicit neural representation.
	[74]	2024	MLP	Design an online framework with a closed-form density model (NASG).
	[105]	2024	K-means	Guide path vertex placement using an irradiance voxel data structure.
Adaptive Sampling	[29]	2011	MLP	Classify different visibility configurations.
	[97]	2018	CNN	Estimate a sampling map to guide sample distributions.
	[155]	2018	CNN	Drive adaptive sampling using an error-prediction module.
	[78]	2020	RL	Train the Q-network with DRL to guide sampling.
	[62]	2020	CNN	Use the sample map to set per-pixel path counts.
	[140]	2022	CNN	Distribute the sampling budget adaptively across regions.
	[43]	2023	CNN	Integrate denoising-awareness into sampling via variance estimation.

Ref. and Archit. are short for reference and architecture, respectively. RL represents reinforcement learning. ED represents encoder-decoder.

2.3 Discussion

These methods are summarized in Table 1. A comparative analysis reveals persistent challenges in sampling techniques and potential research directions:

- **CH1: Insufficient Sample Size.** Inadequate sampling may introduce structured artifacts in rendered images. Solutions could integrate warping transformations with parallax-aware algorithms.
- **CH2: Limitations of Importance Sampling.** Performance degrades when early BSDF-based samples fail to generate representative initial distributions.
- **CH3: View dependent shading effects.** Such effects complicate accurate motion vector calculations, as shading terms may exhibit distinct motion patterns. This issue intensifies in scenes with reflective surfaces.

Incorporating adaptive learning strategies could significantly enhance robustness in the future. Furthermore, combining these two sampling techniques may emerge as a promising area of research.

3 Denoising

MC path tracing produces realistic results using a unified, simple rendering pipeline, including global illumination. However, achieving imperceptible noise levels without post-processing requires thousands of spp, beyond desktop GPUs' capabilities for real-time rendering. Post-processing techniques have been developed to enhance MC denoising and address this challenge by leveraging auxiliary features such as **geometric buffers** (**G-buffers**), samples, and path descriptors (P-buffers). Recent advancements in deep learning have further improved the denoising process by effectively utilizing these auxiliary features to produce high-quality denoised images.

In 2015, Zwicker et al. [186] categorized non-machine learning algorithms into the prior and posterior methods. Our focus here lies predominantly on posterior methods. This approach treats the renderer as a black box, employing a sequence of reconstruction filters while devising error estimates for the reconstruction outcomes. These methodologies were adapted from image denoising techniques, with adjustments to accommodate auxiliary feature buffers. For instance, Bauszat et al. [16] utilized geometric mapping guided filters [63] to reduce noise. Sen et al. [141] introduced the random parameter filtering algorithm to denoise in MC rendering. This method estimated the functional dependency between the sample values and the random parameters used to compute those values. Li et al. [100] incorporated SURE [144] to MC denoising, employing it to estimate the appropriate spatial filter parameters of cross-bilateral filters, with the remaining cross terms' weights hard coding. Building upon this, Rousselle et al. [139] employed SURE to pick from filters with varying weight colors and features from three prospective cross non-local means filters [138]. Moon et al. [120] utilized a linear model to approximate the ground truth and weighted each pixel's error based on auxiliary features. Bitterli et al. [18] constructed collaborative regression using feature buffers. In addition, researchers proposed various algorithms to address the noise in MC rendering. Some of these include new MC formulations with faster convergence [154] and methods that position or reuse samples based on the shape of the multidimensional integrand [59]. Currently, traditional methods for handling complex scenes still rely on a large number of samples to achieve low-noise results, and filtering operations often lead to the blurring of high-frequency details.

We establish key notations before detailing the denoising framework and formalize denoising as a deep learning task. In a typical MC renderer, samples are initially encoded as data vectors x_p (e.g., per-pixel, per-sample, or per-path representations). The objective of MC denoising is to derive a filtered estimate $\hat{c}_p$ that converges to the ground truth $\bar{c}_p$, which corresponds to the asymptotic result under infinite samples. To achieve this, the denoising process operates on a local neighborhood $\mathcal{N}$(p), aggregating and processing a block of data vectors $\boldsymbol{X}_p$ within this region to compute the filtered output at pixel p. The task is formulated as a denoising function $g(\boldsymbol{X}_p; \theta)$, parameterized by θ, which maps the input data block to the final estimate for each pixel p:

$$\hat{\theta}_p = \underset{\theta}{argmin}\, \ell(\bar{c}_p, g(\boldsymbol{X}_p; \theta)), \tag{7}$$

where θ refers to the global parameters of the neural network (e.g., convolutional filters, MLP weights). These parameters are shared across all pixels rather than being assigned individually to each one. The denoised value is $\hat{c}_p = g(\boldsymbol{X}_p; \theta)$ and $\ell(\bar{c}, \hat{c})$ is a loss function between the ground truth value, $\bar{c}$, and the denoised value.

In Equation (7), Bako et al. [11] proposed that function g utilized one of two potential architectures for producing denoised color values: a **direct prediction** convolutional network or a **kernel prediction** convolutional network. Our article focuses on this categorization to organize deep learning-based denoising. Beyond these two methods, there exist several other derivatives as well. Today, the industry offers robust MC denoising solutions, such as NVIDIA's *OptiX AI Denoiser* [131] and Intel's ***Open Image Denoiser*** (**OIDN**) [79]. Both of these advanced methods leverage direct prediction networks to deliver high-quality results.

3.1 Direct Prediction

Applying direct prediction for generating denoised images is a relatively straightforward process. It involves selecting the appropriate size for the network's final layer to ensure that each pixel, p, corresponds to the original color output by the network. Here, L is the index of the last layer in the network, $z_p^L \in \mathbb{R}^3$ represents the denoised color:

$$\hat{c}_p = g_{direct}(\boldsymbol{X}_p; \theta) = z_p^L. \tag{8}$$

Initially, MC denoising primarily revolved around **pixel space**, with $\boldsymbol{X}_p$ in Equation (7). This approach involves regressing the final pixel color values or predicting a filter centered on the pixels. On the other hand, **sample space** related techniques concentrate on filtering a single original sample for each pixel position, showcasing outstanding performance in handling high-energy exceptional value. Nonetheless, these methods often result in increased computational demands and memory usage. **Path space** related exploration emerged in 2021, typically integrating pixels and samples. While path-based approaches offer numerous advantages in complex lighting scenes, their effectiveness diminishes when dealing with more straightforward scenes.

3.1.1 Pixel Space. Chaitanya et al. [23] were the first to apply **recurrent neural network** (**RNN**) [76] technology to MC denoising, achieving interactive rate reconstruction at extremely low sampling budgets. This approach combined skip connections [111] in deep autoencoder structure, marking the initial utilization of these two methods in academic research for light transmission reconstruction problems. End-to-end training in automatic learning enabled more effective incorporation of auxiliary pixel features, such as depth and normals, with no guidance from the user. This method had been integrated into NVIDIA Optix [131] and had undergone several updates. Kuznetsov et al. [97] presented a technique involving joint adaptive sampling and reconstruction to denoise low-sample MC images. Their proposed network comprised two convolutional networks. The second CNN was used to denoise the rendered image and produce the final output. The denoising network employed a CNN-based encoder-decoder architecture. Chaitanya et al. [23] simplified the input by demodulating the noisy RGB images by the albedo of the directly visible material. Their approach stored untextured illumination, removing texture complexity from noisy images, facilitating training, and optimizing network capacity. Therefore, they used a CNN to denoise only the illumination images. In contrast, Kuznetsov et al.'s [97] network involved CNN directly denoising noisy rendered images, enabling handling general distributed effects.

Yang et al. [169] introduced DEMC to encode feature buffers and noisy images with different encoders. The first encoder was a feature buffer for extracting detailed information to enhance image reconstruction in the decoding stage. The second encoder was a **high dynamic range** (**HDR**) [135] image encoder responsible for transforming noisy images into a compact representation of the spatial contexts for the image. Wei et al. [160] developed a technique based on **super-resolution and denoising** (**SRD**) that accelerated the MC rendering by employing an end-to-end architecture. Their approach incorporated deformable convolution to adaptively learn the appropriate receptive field, generating smoother visual effects and keeping more structural details.

Wong et al. [162] introduced a deep residual learning-based [64] method that diverges from traditional indirect learning approaches and estimates filters and kernel weights. Their approach directly mapped noisy input images to their corresponding denoised counterpart. Yang et al. [170] created an end-to-end FRCNN model to integrate feature buffers and directly predict denoised images. They developed an HDR image normalization method to facilitate more efficient and stable model training on HDR images. Meng et al. [114][3] utilized a neural network capable of real-time denoising of 1 spp noisy input images at frame rates of just a few milliseconds per frame. The neural network learned to generate a guide image for splatting noisy data into the grid and then slicing it to read the denoised data. They utilized a differential bilateral grid, called a neural bilateral grid, which enables end-to-end training.

Huo et al. [78] proposed Q-network and R-network. The CNN-based R-network reconstructed the incident radiance field in 4D space. Initially, the R-network was trained, followed by the trained R-network being utilized to prepare the Q-network, aiming at minimizing the loss incurred. Milef et al. [116] presented a method that employed a sequence of small CNNs to denoise at multiple scales. This process involved downsampling the input image and auxiliary features to distinct resolutions, progressively denoising them, and passing the lower resolution as inputs to the next layers. Their denoiser inherently could execute foveated denoising, generating denoised images for varying retinal eccentricity levels that can be combined into a foveated image [54].

[3] https://github.com/xmeng525/RealTimeDenoisingNeuralBilateralGrid

Xu et al. [165][4] were the first to propose using an adversarial learning framework for MC denoising. GANs could enhance denoising networks in generating realistic high-frequency details and global illumination effects by learning from a collection of high-quality MC path tracing images. Their framework leveraged Wasserstein distance [4] to measure perceptual similarity, representing the disparity between denoising and the actual distribution. They demonstrated improved reconstruction of MC integration from multiple samples, resulting in more resilient image outcomes and enhanced processing speeds. Lu et al. [107] designed DMCR-GAN. This method incorporated residual attention networks and hierarchical features modulation of auxiliary buffers. Furthermore, the **residual in residual** (**RIR**) [177] approach was applied with three-level skip connections to ease the flow of information and enable the network to focus on high-frequency information. Lu et al. [106] proposed a **convolution dense block group** (**CDBG**) method, extracting several auxiliary buffers' hierarchical features. They also employed channel and spatial attention mechanisms to emphasize the dependency on channel features and the informative part of spatial features. They enhanced the GAN-based network by introducing the dual residual connection GAN to optimize feature selection and improve potential interactions between modules.

Yu et al. [173][5] were the first to integrate the self-attention mechanism into MC denoising, AFGSA, which capitalized on the non-local means of self-attention, conducting filtering in the embedding space, rendering it well-suited for denoising tasks. Chen et al. [25] introduced a transformer-based [152] denoising network that leveraged multi-scale auxiliary features within a generative adversarial architecture. Their U-shaped denoising network was designed to extract multi-scale texture and geometric features from auxiliaries. The improved transformer module incorporated cross-channel self-attention to capture non-local relationships efficiently with near-linear computational complexity. Their approach demonstrated strong performance in challenging scenes featuring intricate hair and texture details, historically posing difficulties for denoising tasks. Oh et al. [132][6] introduced a transformer-based denoising framework that integrated self-attention mechanisms to tackle this. This approach extracted dual attention scores from colors and auxiliary buffers, enabling it to infer a joint similarity from two distinct input sources with different features.

3.1.2 Sample Space. Işık et al. [81] proposed a method by introducing a pairwise affinity over the features in a pixel neighborhood, from which they assemble dilated spatial kernels to filter the noisy radiance. The denoiser was temporally stable, which could be attributed to two fundamental mechanisms. Firstly, they employed a per-pixel recursive filter with adaptive learning weights to preserve a running average of the noisy radiance and intermediate features. Second, they utilized a small temporal kernel based on the pairwise affinity between features of consecutive frames. Balint et al. [13][7] used a low-pass pyramid filter kernel architecture comprising a lightweight U-Net [136] operating on pixels and encoders operating on samples. Leveraging the pyramid structure, this filter avoided the low-frequency challenges encountered by conventional filters, effectively addressing issues like ringing, blotchiness, and box-shaped artifacts while improving overall image details. Their weight prediction network was trained to partition the input radiance between pyramid layers, to predict kernels for denoising each partitioned and downscaled image, and to guide the upsampling process when combing layers.

[4]https://github.com/mcdenoising/AdvMCDenoise
[5]https://github.com/Aatr0x13/MC-Denoising-via-Auxiliary-Feature-Guided-Self-Attention
[6]https://github.com/CGLab-GIST/joint-self-attetion
[7]https://github.com/balintio/nppd

3.1.3 Path Space. Cho et al. [26][8] introduced a contrastive manifold learning framework [91] to utilize path space features. The framework employed weakly-supervised learning to convert reference pixel colors into dense pseudo labels for light paths. Subsequently, a convolutional path-embedding network induced a low-dimensional manifold of paths by iteratively clustering intra-class embeddings and discriminating inter-class embeddings using gradient descent. The proposed method promoted path space exploration within the reconstructed network by extracting low-dimensional yet meaningful embeddings from the features. Han et al. [61][9] argued that relying solely on geometric buffers and path descriptions may only partially utilize each auxiliary feature. To address this, they proposed a denoising framework emphasizing pixel-wise guidance for using these auxiliary features. Initially, they trained two denoisers, each utilizing different auxiliary features. Subsequently, they developed an integrated network to generate per-pixel ensembling weight maps, which represented pixel-wise guidance for which auxiliary feature should be dominant in reconstructing each individual pixel and used them to ensemble the two denoised results of denoisers.

3.2 Kernel Prediction

The final layer of the network does not directly output the denoised pixel value $\hat{c}_p$. Instead, it predicts a convolutional kernel comprising scalar weights, which are applied to the noisy neighborhood of pixel p to compute $\hat{c}_p$. Let $\mathcal{N}(p)$ denote the $k \times k$ neighborhood centered at p. The output dimensions of the final layer $z_p^L \in \mathbb{R}^{k \times k}$ are designed to match this kernel size. Crucially, the kernel size k is predefined as a hyperparameter prior to training alongside other architectural parameters (e.g., layer width, CNN filter sizes). Notably, the same kernel weights are shared across all RGB color channels.

After flattening z_p^L into a vector where $[z_p^L]_q$ represents its qth entry, we compute the normalized kernel weight via Equation (9) and the denoised pixel color via Equation (10).

$$w_{pq} = \frac{exp([z_p^L]_q)}{\sum_{q' \in \mathcal{N}(p)} exp([z_p^L]_{q'})}. \tag{9}$$

$$\hat{c}_p = g_{weighted}(\boldsymbol{X}_p; \theta) = \sum_{q \in \mathcal{N}(p)} c_q w_{pq}. \tag{10}$$

The kernel weights function as an activation layer applied to the final network outputs across the neighborhood $\mathcal{N}$. This constrains weights to $0 \leq w_{pq} \leq 1$, $\forall q \in \mathcal{N}(p)$, with $\sum_{q \in \mathcal{N}(p)} w_{pq} = 1$. Consequently, the denoised color is guaranteed to lie within the convex hull of its neighborhood in the input image, significantly reducing the solution space compared to direct prediction methods. It ensures that the gradients of the error for the kernel weights are well-behaved. Applying the same reconstruction weights to the independent feature components derived through mathematical or algorithmic decomposition within each layer effectively denoises each layer of a given frame. Generally speaking, kernel prediction trades a larger inductive bias for lower-variance estimates, resulting in faster and more stable training than direct prediction.

Direct prediction denoising may offer more significant power than kernel prediction, as it can directly generate denoised results without explicit filtering. In contrast, kernel prediction is inherently constrained by the limitations of kernel footprints. The magnitude and variability of the random gradient computed during training may be substantial for direct prediction, resulting in

[8] https://github.com/Mephisto405/WCMC
[9] https://github.com/qbhan/GuidanceMCDenoising

a slow convergence speed. The kernel prediction network converges faster with more excellent stability. However, the network design for direct prediction is more straightforward.

3.2.1 Pixel Space. Kalantari et al. [87] discovered an intricate correlation between noisy scene data and the ideal filter parameters and proposed learning this relationship using a non-linear regression model. They integrated a matching filter with an MLP to extract hierarchical features from the local pixel neighborhood. The output was a set of filter parameters, which along with noise samples, were given as inputs to the filter, ultimately generating a filtered pixel. Notably, their method marked the pioneering use of neural networks, or even machine learning, for MC denoising and demonstrated substantial advancements. Their work laid the foundation for kernel predicting denoising networks. Rather than predicting kernel weights directly, they focused on predicting the weights used for the cross-bilateral kernel. This innovative approach sparked the development of learning-based denoising techniques within the research community.

Bako et al. [11] were the first to introduce and employ a kernel prediction method. They utilized CNN to estimate the local weighting kernels to compute each denoised pixel from its neighbors. The trained network had the potential to generate production data and could be effectively applied in a variety of scenes. Additionally, they innovatively presented a two-network framework for denoising, including diffuse and specular components of the images separately, along with a normalization procedure that notably enhanced the restoration for images with HDR. Vogels et al. [155] expanded the capabilities of kernel prediction networks by combining them with several task-specific modules and optimizing the assembly using an asymmetric loss. The source-aware encoder, the first module in the assembly, extracted low-level features and integrated them into a common feature space, enabling quick adaptation of a trained network to novel data. The complete network was trained using a class of asymmetric loss functions designed to preserve details and was designed to give the user direct control over the variance-bias tradeoff during inference.

Fan et al. [40][10] treated a neural network as an encoder to produce a compact single-channel format of the filtering weight, depicted as importance maps. These importance maps were fed into a hand-crafted decoder to generate the complete kernel map. Notably, they adopted a variant of the RepVGG Block [34] structure in the prediction phase to create ImportanceNet. The RepVGG Block structure could improve the capacity of a fully CNN during training and would not introduce additional time costs to network inference. Hasselgren et al. [62] introduced a technique that optimized sampling and denoising. Their network consisted of a sample map estimator and a denoiser. We focused solely on the denoising network here. They used a small U-Net architecture with hierarchical kernel prediction [155] to create a robust and efficient denoiser.

Zhang et al. [175] designed an architecture integrating a pixel-based neural decomposition module with a kernel prediction denoiser. The neural decomposition module learned to predict noisy components and corresponding feature maps, which were then continuously reconstructed by the denoising module. The components were predicted based on statistics aggregated at the pixel level by the renderer. By denoising these components separately, each component's kernel could be tailored to adapt to the noise signal characteristics of that specific component. The U-Net network was utilized in the decomposition module and denoising module. Fu et al. [44] implemented a sparse auxiliary feature encoder to streamline U-shaped kernel prediction networks. With sparse convolutions, the computational complexity of the auxiliary feature encoder was reduced by 50-70% without apparent performance drops. The denoising architecture introduced by Salehi et al. [140] featured a two-pass kernel predictive denoiser with a U-Net architecture similar to [175]. Thomas et al. [151] adopted a neural network architecture that integrated supersampling and denoising,

[10] https://github.com/Rendering-at-ZJU/weight-sharing-kernel-prediction-denoising

effectively lowering costs compared to two separate networks. The computationally heavy U-Net-based feature extractor was set to use 8-bit precision and ran at a lower resolution, aiding in faster execution.

3.2.2 Sample Space. Designing a network architecture that effectively learns the mapping between samples and images presents significant challenges: the arbitrary order of samples and how they should be treated necessitates processing them in a permutation invariant manner. Gharbi et al. [48][11] developed a kernel prediction architecture that split individual samples onto nearby pixels to address this. Based on the U-Net architecture, their network was designed to be perfectly invariant to per-pixel permutations of the samples, leveraging two key concepts: individual sample embeddings and per-pixel context features. The sample embedding was a non-linear feature space associated with each sample that was ultimately responsible for predicting the scattering kernel to nearby pixels. Munkberg et al.'s [124][12] approach involved utilizing data from samples partitioned into layers and filtered individually, enabling the network to handle outliers and complex visibility effectively. Finally, the layers were reconstructed via front-to-back alpha blending. The computational expense of denoising on the per-sample was typically high and escalated with improvements in input quality. Their study used a data-driven approach and allowed a neural network learn this representation to achieve effective smoothing while balancing cost and quality.

3.2.3 Path Space. Lin et al. [102] introduced a network that integrated light transport covariance from path space and an improved loss function for MC denoising. This network separated the feature buffers from the color buffer to enhance the detail effects, extracting features independently and merging them into the shallow kernel predictor. The pre-trained network utilized VGG-19 [143], as VGG-19 could provide high-dimensional feature information for an image. In addition, the path space light transport covariance feature contributed to detail preservation. Stemming from Durand et al.'s [39] light transport frequency analysis framework, they computed the frequency content of the local light field around a given light. Lin et al. [103] concurrently employed three-scale features–pixel, sample, and path–to preserve sharp details. They designed a hybrid feature extractor based on Res2Net [47] and U-Net, along with a pixel feature attention mechanism to enhance the impact of smooth features. The network output the filter kernels for each path and computed the final denoised pixel radiance by a splatting operation.

3.3 Other Techniques

Apart from direct prediction convolutional networks and kernel prediction convolutional networks, there are also some MC path tracing denoising methods based on deep learning. These methods involve using neural networks to predict filter parameters based on traditional filters, utilizing the gradient-domain techniques, combining images before and after post-denoising (e.g., post-correction), being specific to volume renderings, and implementing framework enhancements for denoising.

3.3.1 Filter Parameters. Xing et al. [164] employed SURE to estimate the noise level for each pixel, guiding the adaptive sampling. Subsequently, they derived features from the outcomes of adaptive sampling for denoising purposes. A modified MLP network was utilized during the reconstruction stage to capture the complex relationship between the extracted features and the optimal reconstruction parameters. They reconstructed the final image using an anisotropic [112] filter with the parameters predicted by neural networks.

[11] https://github.com/adobe/sbmc
[12] https://github.com/NVlabs/layerdenoise/

3.3.2 Gradient-domain Rendering. Gradient-domain light transport represents a recent technique capable of significantly enhancing MC rendering speed, leveraging a gradient-based estimation and reformulating the rendering problem as an image reconstruction [71]. Typically, this method involves estimating pixel intensities alongside the image gradient and utilizing image reconstruction to synthesize the final image. In addition to estimating pixel intensities, they compute MC estimates of image gradients using correlated samples. They combined these two samples into the final image by solving a screened Poisson problem. Guo et al. [56][13] introduced an innovative multi-branch autoencoder, termed GradNet, designed end-to-end to learn directly from the image gradients of noisy input images and produce low variance, high-quality images with U-Net networks. They devised and trained the network in a completely unsupervised manner, learning directly from the input data, which marked the first solution of integrating unsupervised deep learning into gradient-domain rendering frameworks. Kettunen et al. [94][14] created a hybrid of **Densely Connected Convolutional Networks (DenseNets)** [73] and U-Net, presenting a novel gradient-domain rendering reconstruction approach. Their dense variant of the U-Net autoencoder replaced the screened Poisson solver of the previous gradient-domain technique, using an auxiliary feature buffer as input. The network was optimized to minimize the perceptual image distance metric calibrated to the human visual system.

3.3.3 Post-correction. Despite the rapid advancements in existing deep learning-based MC denoising techniques, they continue to grapple with residual noise specific to methods or system errors. Back et al. [6][15] proposed a unified framework that leveraged a stack of images–one with independent estimates and the other with corresponding correlated estimates. Correlated pixel estimates were derived from various existing methods, such as denoising and gradient-domain rendering. Subsequently, the framework combined the two images via a combination kernel, represented as a weighting function with a deep neural network that exploited the correlation among pixel estimates. This approach marked the first instance of addressing structured noise and effectively denoising inherent in current methods.

Firmino et al. [42][16] proposed a progressive denoising technique that employed neural networks to find per-pixel mixing parameters. In this approach, one image was an MC rendering image, while the other represented a denoised image. Error estimations were conducted using the estimated variance of the sample mean for the former and SURE for the latter. However, these methods might falter when the rendered image (i.e., the test data) had different characteristics from the images in the training dataset. In response to this challenge, Back et al. [7][17] proposed a self-supervised framework that post-corrects the results of a supervised learning model using their input and output without any pretraining. Their work marked the first application of self-supervised learning in MC rendering denoising. In contrast to supervised learning approaches, their model did not need a reference image in its training process, allowing noisy test images to self-correct the model and improve denoising performance. MC denoising based on supervised learning improved denoising quality but inevitably introduced systematic errors, i.e., the denoising bias. Unlike other types of errors, variance, denoising bias did not decline rapidly as the number of samples increased, which could technically lead to slow numerical convergence of denoising the techniques. Gu et al. [53][18] proposed a new combination framework based on the **James Stein (JS)** estimator [83], which

[13] https://github.com/guojienju/GradNet
[14] https://github.com/mkettune/ngpt
[15] https://github.com/CGLab-GIST/deep-combiner
[16] https://github.com/ArthurFirmino/progressive-denoising
[17] https://github.com/CGLab-GIST/self-supervised-post-corr
[18] https://github.com/CGLab-GIST/NeuralJS_combiner

combined a pair of unbiased and biased rendered images, e.g., a path tracing image and its denoising results. Unlike post-correction techniques, their framework helped input denoisers have lower errors than their unbiased inputs without relying on accurate estimates of denoising errors per pixel.

3.3.4 Volume Rendering. The field of volume MC rendering denoising emerged around 2020. The primary algorithm for producing volumetric effects is volumetric path tracing based on the **radiative transfer equation** (**RTE**) [178]. This method involved tracing light paths as they scattered and got absorbed within participating media. Rendering scenes with volumetric effects often involved more scattering events per path than scenes with only surface interactions, leading to increased noise levels or longer rendering times. As a result, efforts have been made to speed up volumetric rendering, including techniques such as various forms of **photon mapping** (**PM**) [17], enhancements in free flight distance sampling [127], transmittance estimation [32]. Interested readers are encouraged to explore surveys conducted by Cerezo et al. [22] and Novák et al. [126] for a comprehensive overview of available volume rendering techniques.

Hofmann et al. [69][19] extended the application of machine learning techniques, previously limited to surface MC denoising, to path-traced visualizations of medical volumetric data. Their approach proposed a learning-based method for denoising path-traced videos of volumetric data by increasing the sample count per pixel through spatial (integrating neighboring samples) and temporal filtering (reusing samples over time). Xu et al. [166] introduced the first deep learning approach for reconstructing images in volumetric rendering gradient-domain density estimation. Their method combined a two-scale network of U-Net and an encoding shift connection module, which effectively smoothed low-frequency areas while preserving high-frequency details. They implemented three branches within the network, like GradNet [56], separating the features from color images.

Zhang et al. [176] adopted the concept of the volumetric feature sets, aiming at enhancing the performance of advanced denoisers in volumetric rendering. They designed a pipeline to select the most beneficial auxiliary feature set by using a probe denoiser with feature dropout to alleviate the need for training multiple denoisers during feature selection. Their method for constructing the feature set considered the importance of other features, paving the way for a more systematic exploration of how alternative feature selection approaches impact various tasks. Jabbireddy et al. [82] integrated deep learning denoising techniques with volume rendering based on neural bilateral grids [114], replacing the heavy pipeline of MC volume rendering and achieving high-quality rendering at interactive rates. Their approach involved volumetric path tracing to render noisy images using just 1 spp. The noisy image and the proposed volume auxiliary features were passed through the neural bilateral grid denoiser to obtain a high-quality result.

Hofmann et al. [68] utilized a denoising architecture that built upon their prior research [69]. Denoising techniques inspired their approach [124], where the signal was partitioned into volume and surface components. Using the temporal denoiser with kernel predicting neural network [11, 62], they denoised these components individually and finally composited the denoised signal based on volume transmittance. Their methodology could generate temporally stable results even with low sample counts (4 spp) at interactive rates. Pixar Animation Studio was also actively engaged in denoising volumetric rendering. Their research [185] involved modifying the production renderer to generate potential features that could enhance denoising quality. Subsequently, they employed a state-of-the-art feature selection algorithm to detect the most effective combination.

[19] https://github.com/nihofm/ndptmvd

This process significantly enhanced denoised image quality, recovering more details and removing blotchiness more effectively.

Participating media is a ubiquitous material observed widely in the real world, such as clouds, milk, jade, skin, etc. Correctly handling the interaction between these materials and light is pivotal for elevating rendering realism. Hu et al. [70] introduced a practical framework with a light-weight feature fusion neural network, the **Multi-feature Radiation Prediction Neural Network** (**MRPNN**), for rendering real-time high-order scattered radiation of participating media. Their innovation included reformulating the RTE, presenting transmittance fields, and generating a low cost as auxiliary information to help the network better approximate the RTE, significantly reducing the size of the neural network. Additionally, they proposed a frequency-sensitive stencil design to handle non-cloud shapes, resulting in accurate shadow boundaries.

3.3.5 Framework Enhancements for Denoising. This approach primarily involves systematically optimizing the denoising process to boost traditional methods' robustness, efficiency, and generalization. This is achieved through improvements in the architecture design, feature interaction mechanisms, or learning paradigms of denoising algorithms. Unlike the direct denoising techniques discussed earlier, this approach focuses on the modular enhancement of denoising frameworks rather than developing independent new denoising operators.

Zhang et al. [175] proposed an innovative approach by integrating a pixel-based neural decomposition module with a kernel prediction denoiser. This decomposition module, a neural network, effectively decomposed noisy rendered images into individual components, each of which could be denoised separately. In addition to processing noisy color images, the module could decompose user-defined components commonly generated by the renderer to enhance the denoising performance. Cho et al. [26] introduced a manifold learning framework to enable MC reconstruction to fully exploit path information, including first bounces and multi-bounce features. This approach aimed at extracting compact and useful embeddings from high-dimensional path features, effectively compensating for the sparsity in the path space and enhancing reconstruction performance. Zhang et al. [176] introduced an innovative feature set for volume denoising and an algorithm to identify the most compelling feature combinations. By evaluating probe denoising across different feature sets and measuring the resulting quality differences, they predicted the impact of each feature combination on denoising quality.

3.4 Discussion

Table 2 summarizes various approaches with distinct objectives and evaluations on diverse datasets, complicating a generalized performance analysis. Nevertheless, critical observations from these designs include: (i) most methods operate in supervised or semi-supervised modes; (ii) CNN architectures are predominantly adopted; (iii) evolving deep learning technologies enhance denoising quality and processing efficiency within similar network types. The rendering quality is usually proportional to sample counts and time. Integrating denoising with other technologies (e.g., sampling techniques or upsampling) has become a key development trend.

MC noise exhibits distinct characteristics that complicate denoising: (i) Spatially, noise manifests as high-frequency artifacts in regions with complex light transport (e.g., caustics) and low-frequency bias in diffuse interreflections [11]. (ii) Temporally, uncorrelated noise across frames leads to flickering, requiring temporal accumulation or motion compensation [163]. (iii) Scene-dependent factors such as material roughness and light path depth further modulate noise patterns [124].

Over the past decade, denoising technology has made significant strides, primarily propelled by advancements in deep learning capabilities. However, this field continues to confront several challenges.

Table 2. Methods for Denoising Techniques

Method	Ref.	Year	Archit.	Description
RAE [23]	3.1.1	2017	RNN/ED	Combine a skip connection in deep autoencoder structure.
DASR [97]	3.1.1	2018	CNN/ED	Divide into two CNNs: estimate a sampling map and produce denoised images.
DEMC [169]	3.1.1	2019	CNN/ED	Propose a Dual-Encoder network with a feature fusion subnetwork.
[160]	3.1.1	2021	CNN/ED	Propose an end-to-end network that first upsamples and then denoises.
[162]	3.1.1	2018	CNN	Propose a deep residual learning-based method using three auxiliary features.
FRCNN [170]	3.1.1	2019	CNN	Devise an HDR image normalization method to train the model.
[114]	3.1.1	2020	CNN	Use differentiable neural bilateral girds, allowing end-to-end training.
[78]	3.1.1	2020	CNN	Reconstruct the incident radiance field using a CNN-based R-network.
[116]	3.1.1	2021	CNN	Generate images for varying retinal eccentricity levels.
MCGAN [165]	3.1.1	2019	GAN	Present the first adversarial learning framework for MC denoising.
DMCR-GAN [107]	3.1.1	2020	GAN	Propose an adversarial approach with attention networks.
DuRCGAN [106]	3.1.1	2021	GAN	Devise a GAN based on dual residual connections.
AFGSA [173]	3.1.1	2021	Transformer	Stack self-attention blocks and combine them with a transformed block output.
MSAFFT [25]	3.1.1	2023	Transformer	Present a transformer-based network guided by auxiliary features.
[132]	3.1.1	2024	Transformer	Design a transformer block with joint self-attention.
[81]	3.1.2	2021	CNN	Employ a per-pixel recursive filter with adaptive learning weights.
[13]	3.1.2	2023	CNN	Present a kernel architecture aligned with low-pass pyramid filters.
[26]	3.1.3	2021	CNN	Design a manifold learning method that applies to pixel and path space.
[61]	3.1.3	2023	CNN	Employ two denoisers trained on pixel and path space for denoising.
[87]	3.2.1	2015	MLP	Employ an MLP to predict the weights used for the cross-bilateral kernel.
KPCN [11]	3.2.1	2017	CNN	Present a two-network framework including diffuse and specular components.
[155]	3.2.1	2018	CNN	Combine kernel prediction networks with multiple task-specific modules.
[40]	3.2.1	2021	CNN	Learn to predict an encoding of the filtering kernel weights.
[62]	3.2.1	2020	CNN	Use a small U-net architecture with hierarchical kernel prediction.
[175]	3.2.1	2021	CNN	Decompose noisy MC renderings automatically into components for the denoiser.
[44]	3.2.1	2021	CNN	Facilitate the kernel prediction network with a sparse auxiliary feature encoder.
[140]	3.2.1	2022	CNN	Sample training data by using a few per-pixel statistics of the target distribution.
[151]	3.2.1	2022	CNN	Introduce a network architecture that combines supersampling and denoising.
SBMC [48]	3.2.2	2019	CNN	Present the first CNN that learns to denoise MC renderings directly from samples.
LBMC [124]	3.2.2	2020	CNN	Present an architecture that learns to partition samples into layers.
[102]	3.2.3	2020	CNN	Propose a new input feature: light transport covariance from path space.
[103]	3.2.3	2020	CNN	The network handles three-scale features: pixel, sample, and path.
[164]	3.3.1	2020	MLP	Use MLPs to predict parameters and cross-bilateral filters to denoise.
GradNet [56]	3.3.2	2019	CNN/ED	Design the first unsupervised network to enhance MC rendering.
NGPT [94]	3.3.2	2019	CNN/ED	Use the U-Net autoencoder to replace the screened Poisson solver.
DC [6]	3.3.3	2020	CNN	Combine two independently denoised images via a combination kernel.
[42]	3.3.3	2022	CNN/ED	Use SURE to estimate error and infer a per-pixel mixing parameter.
[7]	3.3.3	2022	CNN	Propose a self-supervised loss that can guide the network without reference.
[53]	3.3.3	2022	CNN/ED	Present a framework built upon a JS estimator that merges a pair of images.
NDPT [69]	3.3.4	2020	CNN/ED	Use additional features and a loss function for volumetric cases.
[166]	3.3.4	2020	CNN/ED	Design a network for volumetric photon density estimation reconstruction.
[176]	3.3.4	2022	CNN/ED	Propose a method for feature sets for MC renderings with volumetric content.
VDN [82]	3.3.4	2023	CNN	Combine learning-based denoising with direct volumetric rendering.
[68]	3.3.4	2023	CNN	Combine volume and surface denoising in real-time.

— **CH1: Specular objects.** Specular light paths describe the trajectory of light as it travels exclusively through reflection or transmission within a scene. A defining characteristic of this optical path is that the light's direction follows strict physical laws. Processing specular objects using deep learning techniques presents considerable challenges. However, exploring path-based denoising methods and leveraging temporally reliable motion vectors holds promising potential for tackling these challenges in future research.
— **CH2: Training dataset.** An insufficient dataset leads to unexpected outcomes when networks encounter scenarios not covered during training. Consequently, challenges such as fog, motion blur, depth of field, and other distributed effects remain unsolved. Munkberg et al.'s deep-wise layer decomposition [124] technique shows promise in addressing these limitations.
— **CH3: Ghosting artifacts.** Unsightly ghosting artifacts can occur when specific networks inaccurately reuse temporal information. This issue is typically more noticeable in static

images than in motion sequences. Occasionally, artifacts near specular regions can be particularly prominent because of scalability issues related to sample counts and computational efficiency.
— **CH4: Over-blurring.** This occurs when certain filters struggle to reconstruct a clean, detailed image of the available samples, necessitating a tradeoff with artifacts. While perceptual loss functions can partly guide the learning process, denoising networks occasionally yield unsatisfactory outcomes. Incorporating multi-scale structures can mitigate this issue.

Deep learning-based denoising technology, serving as a post-processing method for MC path tracing, demonstrates superior denoising capabilities compared to traditional techniques. Denoising technology research has evolved from focusing solely on individual pixels to encompassing path and sample spaces, resulting in significant advancements in detail reconstruction. This additional dimension of rendering challenges has garnered substantial academic interest, potentially shaping the future landscape of next-generation generative rendering applications and real-time path tracing.

4 Upsampling

As the demand for high resolution, high refresh rate, and realistic visuals continues to rise, the computational workload of real-time rendering has significantly amplified, often overwhelming the capabilities of most graphics cards. One of the most popular solutions to address this challenge is to render images at a lower resolution to reduce the rendering overhead. Subsequently, efforts are made to accurately upscale the low-resolution rendered image to the target resolution, balancing performance and visual fidelity.

While deep learning techniques have seen success in upsampling photographic images and videos, they often need to be more suitable for rendering content in real-time rendering. The pixels in a photograph represent area integration based on physical sensing, capturing the scene's essence. In contrast, pixels are initially treated as points in screen space during rendering. Directly sampling from a single point can lead to aliasing, distorting the final image. The renderer's pixel value calculation is essentially an approximation of MC integration, where continuous scenes are mapped to discrete pixels by blending random sampling with filtering techniques, offering a refined approximation of the original visual data. Consequently, the rendered content is typically highly aliased, particularly at a lower resolution. It makes upsampling for rendered content an antialiasing and interpolation challenge, differing from existing upsampling works in the computer vision community focused on the deblurring problem. On the other hand, pixel samples in real-time rendering are accurate, and, more importantly, motion vectors are available nearly for free at subpixel precision. These factors introduce new opportunities and complexities to the upsampling problem in rendering, expanding the repertoire of rendering-centric deep learning techniques. Existing deep learning-based upsampling techniques for image rendering can be broadly categorized into two types. The first type is **spatial upsampling**, which increases the resolution of images from one level to another. The second type is **temporal reconstruction**, which involves inserting generated frames between existing frames to enhance rendering quality.

4.1 Spatial Upsampling

In recent years, extensive research has been into learning-based video super-resolution [3, 158]. Despite sharing similar goals with rendered image upsampling, the two tasks differ in several key aspects. Firstly, while the super-resolution of input videos often involves aliasing-free and low-pass filtering, image processing typically employs point-sampled and aliased. Secondly, video super-resolution usually takes full GPU resources, whereas real-time rendering upsampling requires less computational and memory I/O budget. Additionally, unlike video capture, which lacks the accurate underlying scene and motion information, the rendered frames often benefit from accurate motion vectors and depth information of the scene.

According to the various image priors, Yang et al. [167] categorized single-image super-resolution into several approaches: prediction models, edge-based methods, image statistical methods, and patch-based methods. Prediction models generate **high-resolution** (**HR**) images from **low-resolution** (**LR**) inputs using predefined mathematical formulas without training data. Examples of such methods include bilinear, bicubic, and Lanczos. The IP method [80] exploited a predefined downsampling model from a HR image to a LR image. Edge-based methods focus on leveraging edge features, such as edge depth, width, or gradient profile [146], to learn priors that aid in reconstructing HR images. Image statistical methods rely on various image characteristics as priors to predict HR images from LR inputs. For instance, some methods utilize heavy-tailed gradient distributions [75], which help reduce computational load. Block-based methods involve using pairs of LR and HR training images, from which patches are extracted to learn the mapping function. For example, Chang et al. [24] employed an external dataset for this approach.

NVIDIA **Deep Learning SuperSampling** (**DLSS**) [130] is a neural graphics technology that multiplies performance using AI by Tensor Core to create entirely new frames and display higher resolution through image reconstruction–all while delivering best-in-class image quality and responsiveness. Intel **Xe Super Sampling** (**XeSS**) [67] takes our gaming experience to the next level with AI-enhanced upscaling, enabling more performance with high image fidelity. Revel in ultra-high-definition visuals with high performance powered by hardware acceleration and an AI-based algorithm. And the solution that does not rely on machine learning, such as AMD **FidelityFX Super Resolution** (**FSR**) [1]. However, these resources lack open-source materials and detailed implementation guidance, thus failing to offer us valuable references for future research.

Xiao et al. [163] introduced a specialized temporal neural network tailored for upsampling rendered content, leveraging diverse rendering attributes such as color, depth, and motion vectors while optimized for real-time usage. Their method achieved 4×4 upsampling with remarkable spatiotemporal fidelity, enabling high-quality aliasing super-resolution processing for rendered video content. Their approach innovatively manipulated frame distortion within the target (high) resolution space rather than the input (low) resolution space to maximize the utilization of rendering data details like color sampling and sub-pixel precise motion vectors. It involved projecting input pixels into high-resolution space via zero upsampling before warping. They adopted a weighted combination of perceptual loss and structural similarity index for training losses, computed using the VGG-16 layer network [85].

Wei et al. [160] presented a network that joint SRD post-processing within MC path tracing, employing an end-to-end network architecture. They adopted the classic network EDSR [101] for super-resolution. Moreover, they concluded that super-resolution requires a smaller receptive field, whereas denoising requires a larger one. Thomas et al. [151] implemented upsampling following denoising, opting to upsample the composite image rather than singular signals. This approach reduced the necessity for numerous filtering operations at the lower target resolution and performed better at reconstructing high-frequency details. Generally, upsampling relies on learning-based methods employing end-to-end training of a "black-box" neural network. However, Vouga et al. [156] devised a "white-box" solution wherein pixels were categorized into different categories, generating upsampling results based on this classification. This approach stemmed from recognizing that the fundamental challenge in rendering time oversampling lies in distinguishing the pixels comprising occlusion, aliasing, or shading changes. Although samples from these pixels might display similar temporal radiance change, they necessitated different strategies to yield accurate upsampling outcomes. Building on this insight, they blended the current frame with the warped last frame via a learned weight map to obtain the upsampling results. Each step involved compact neural networks, and tailored loss functions were crafted for pixels belonging to various categories. Because of the limitation between frame upsampling and extrapolation, quality and latency problems often arise.

Yang et al. [168] proposed an efficient **mobile neural supersampling** (**MNSS**) for real-time rendering applications on hardware-limited platforms like smartphones. Their approach involved incorporating alternative sub-pixel sampling models during the rasterization process, allowing for the creation of a relatively small reconstruction model that still delivered high image quality. They implemented a mechanism to accumulate new samples into a high-resolution history buffer and introduced effective history check and re-usage schemes to enhance temporal stability. By doing so, they achieved a significant milestone in advancing real-time neural supersampling research tailored to mobile devices. Zhong et al. [180][20] employed a low-cost, high-resolution auxiliary G-buffer as an additional input to predict high-quality upsampling reconstructions. To utilize material information from the G-buffer, they utilized pre-integrated BRDF demodulation to effectively filter out HR details and transform the color frame **super-resolution** (**SR**) task into a more easily demodulated irradiance SR problem. They proposed the H-Net architecture employing pixel shuffling operations to effectively and efficiently align and fuse HR and LR information.

4.2 Temporal Reconstruction

Interpolation-based frame rate upsampling is widely used for temporally upsampling videos [9]. Typically, these methods are based on computing the forward and backward optical of the last couple of frames in the video sequences, subsequently, by leveraging optical flow to reproject the future and past frames to estimate the intermediate frames. While recent years have witnessed significant advancements in deep learning-based video interpolation methods [14, 125] to enhance the flow prediction quality and warping occlusion. There are two evident shortcomings in transferring these video processing techniques to real-time rendering. Firstly, they struggle to address aliasing issues effectively. The aliasing present in rendered content stems from low sampling rates, whereas pixels in photographic images are formed by integrating light rays. Secondly, there is high processing latency. Video methods primarily focus on interpolation rather than extrapolation, necessitating future frames (T+1) for generating the current frame (T). Consequently, these methods unavoidably introduce additional display delays. Moreover, contemporary rendering techniques often leverage G-buffers and motion vectors to enhance rendering and upsampling quality.

Dong et al. [38] conducted an insightful survey of video frame interpolation. Before 2010, most algorithms focused on motion-compensated, typically involving two key stages: motion estimation and motion-compensated frame interpolation. Ha et al. [58] introduced a motion-compensated frame interpolation algorithm tailored for new block-based motion estimation. Kang et al. [90] presented an adaptive motion vector smoothing to smooth motion vectors considering true neighboring motion vectors. Huang et al. [72] developed a correlation-based motion vector processing method to detect and correct unreliable motion vectors, effectively preventing visual errors. With the advancement of optical flow estimation methods, subsequent research shifted toward utilizing optical flow between input frames. Welberger et al. [161] proposed the TV-L1 denoising algorithm, driven by optical flow, to address video interpolation challenges.

Guo et al. [55] presented ExtraNet,[21] a neural network that predicts accurate shading results in image space based on previously rendered frames. To achieve optimal image quality in the extrapolated frames, they employed G-buffers from Unreal Engine 4 [45]. Furthermore, they adopt gated convolutions [172] in the network, providing a learnable feature selection mechanism. The

[20] https://github.com/Isaac-Paradox/FuseSR
[21] https://github.com/fuxihao66/ExtraNet

Table 3. Methods for Upsampling Techniques

	Ref.	Year	Archit.	Description
Spatial Upsampling	[163]	2020	CNN	Introduce a temporal neural network with rendering attributes.
	[160]	2021	CNN	Propose an end-to-end network: upsampling first, then denoise.
	[151]	2022	CNN	Design a loss function combining denoising and upsampling.
	[156]	2022	ED	Divide into two subnetworks: pixel classification and image composition.
	[180]	2023	CNN	Implement pre-integrated BRDF demodulation and H-Net.
Temporal Reconstruction	[55]	2021	CNN	Perform irradiance in-painting and ghosting-free shading prediction.
	[20]	2021	CNN	Estimate nonlinear motion via auxiliary features.
	[21]	2023	CNN	Predict kernels for image splat processing.
	[65]	2023	CNN	Treat aliasing and occlusions as reshading zones.

tasks were divided into irradiance in-painting for regions that cannot find historical correspondences and accurate ghosting-free shading prediction for regions where temporal information is available. He et al. [65] presented a novel framework, **Space-time Supersampling** (**STSS**), for real-time rendering, exploiting the common context and mechanism of frame extrapolation and supersampling. The framework streamlines the workflow by adopting a unified context and a shared neural network. It only requires 4 ms per frame at the 1080P setting. They proposed a novel reshaping mechanism that uniformly addresses aliasing and warping holes as reshaping regions and presented the **Random Reshading Masking** (**RRM**) and **Efficient Reshading Module** (**ERM**) for compensating the aliasing and warping holes.

Briedis et al. [20] introduced a method tailored for high-quality frame interpolation for rendered content. This approach leverages the renderer's auxiliary feature buffers to estimate non-linear motion between keyframes, which is preferable over using rendered motion vectors in cases where those are available. Improvements, occlusion handling, and compositing enhancements yield production quality results across various scenarios. Following that, Briedis et al. [21] enhanced rendering robustness by introducing an attention-inspired kernel prediction method and proposed a technique for estimating the interpolated frame by predicting spatial varying kernels that operate on image splats. This method consisted of two key features. First, a kernel-based frame synthesis model anticipates the interpolated frames as a linear mapping of the input images. Consequently, all layers can be interpolated, and a different composition can be created without additional steps. The second feature involved an adaptive interpolation scheme, which optimizes the decision to render or interpolate image regions within a given frame sequence: frame interpolation should be for optimal outcomes in image regions, while rendering is applied to more challenging image patches.

4.3 Discussion

Table 3 summarizes representative upsampling methods. Driven by advances in deep learning, rendering-based upsampling techniques have evolved rapidly, yet they now confront critical challenges:

- **CH1: Input scene dependency.** When scene regions exhibit severe spatial aliasing or insufficient frame information, upsampling fails to reconstruct missing details.
- **CH2: Stability-sharpness tradeoff.** Loss function weighting directly impacts this balance. Determining optimal weights remains a key challenge.
- **CH3: Limited physical consistency.** Models may violate physical laws, yielding implausible outputs or deviations from ground truth.

As deep learning-based path tracing upsampling techniques progress, practical robustness is improving. Future methods are expected to reduce data dependence, deploy lightweight models for

Table 4. The Classification of Development Trends

Type	Studies	Description
Tradition Approaches	[110], [52], [19], [145], [8]	Improve rendering quality under limited computing resources without relying on data training.
Neural Rendering	[50], [51]	Create a new image by "rendering" the input image within a particular scene.
Specialized Rendering	[88], [134], [181]	Specialized networks are designed to render intricate objects or handle complex conditions.
NeRF	[115], [98], [57], [118], [46]	Reconstruct intricate 3D scenes with rich details and dynamic perspective shifts.
3DGS	[93], [109], [41], [108]	Offer an explicit scene representation and view synthesis without reliance on neural networks.
Radiance Caching	[123], [84]	Reconstruct the incident radiance field following denoising.
Photon Mapping	[174], [27]	Integrate photon mapping techniques for enhanced denoising.

real-time mobile applications, and enforce consistency to enhance user interaction and physical accuracy.

5 Development Trends

Over the past decade, MC rendering has seen remarkable progress, with deep learning techniques unveiling powerful new capabilities. Yet, the scope of research extends far beyond these advances. Alongside sampling, denoising, and upsampling developments, various related studies offer valuable insights that can further illuminate MC path tracing. Table 4 shows the classification of development trends.

5.1 Traditional Approaches

Although deep learning-based rendering techniques offer notable advantages in complex light field reconstruction and real-time rendering, owing to their powerful nonlinear mapping capabilities and data-driven nature, research methods based on MC path tracing continue to possess irreplaceable academic value and practical significance.

Manzi et al. [110] utilized gradient-domain rendering techniques to sample finite differences over the image plane, introducing temporal finite differences. They then formulated a corresponding 3D spatiotemporal screened Poisson reconstruction problem, solving it simultaneously over windowed batches of multiple frames. Gruson et al. [52][22] introduced the first image-spatial gradient-domain formulation for rendering in participating media, developing four new rendering algorithms. Bitterli et al. [19] introduced a direct lighting dream discount method based on resampling importance sampling. This technique enabled the unbiased reuse of adjacent samples in both space and time, leading to more effective management of biased variables. Su et al. [145][23] introduced a subspace-based importance sampling method based on the concept of probability-connected subspaces. This approach first sampled the photon space and then resamples the photon paths within it, significantly enhancing rendering quality while maintaining minimal computational complexity. Back et al. [8][24] introduced a denoising strategy based on input-dependent kernels, effectively reducing noise in input pixel estimates while minimizing denoising bias. Their article demonstrated that these input-dependent kernels could produce uncorrelated input pixel estimates, thanks to uncorrelated statistical theory.

[22] https://github.com/gradientpm/gvpm
[23] https://github.com/ssufujia/SPCBPT-Code
[24] https://github.com/CGLab-GIST/input-dependent-uncorrelated-weighting

5.2 Neural Rendering

Neural rendering enables the controllable and high-quality synthesis of new images from input images/videos. A typical neural rendering approach takes input images corresponding to certain scene conditions (viewpoint, lighting, layout, etc.). It then builds a **"neural"** scene representation from this input and **"renders"** it under novel scene properties to synthesize novel images.

Granskog et al. [50] adopted an adaptively partitioning neural scene representations approach. This method effectively disentangled lighting, material, and geometric information. Such a design preserved the orthogonality of these components, improved the interpretability of the model, and allowed the compositing of new scenes by mixing existing components. In 2021, they introduced a neural scene graph [51], a modular and controllable representation of scenes with elements learned from data. It was constructed from elements gleaned from data supplied by the user and references learning elements. These elements encapsulated various scene objects and material definitions, and constituted the leaf nodes of the structure, and were stored as high-dimensional vectors. Artists could easily manipulate the position and appearance of scene objects using familiar transformations, with support for non-linear transformations. Despite its initial visual simplicity, integrating traditional editing methods with learned representations marked a significant stride toward achieving controllable neural rendering.

Unlike traditional MC path tracing, neural rendering offers a novel approach to scene representation, replacing conventional grids and textures. This shift enables more efficient scene modeling, alleviating the computational burden of complex environments and significantly enhancing rendering performance.

5.3 Specialized Rendering

Complex scenes, such as clouds, skies, and luminaires, incur high rendering costs, especially when handling challenging, complex refractive geometry that demands extensive sampling via MC approaches for evaluation. Previous attempts to speed up this process have often needed to be more accurate, involved storing large lightfields, or did not fit well into modern path tracing frameworks. Inspired by the success of deep learning, recent studies have explored the application of machine learning frameworks tailored for rendering tasks.

Kallweit et al. [88] designed **Radiance-Predicting Neural Networks** (**RPNN**), an effective method for generating atmospheric cloud images, combining MC integration and neural networks. They utilized MLPs to pre-learn spatial and directional radiation flux distribution from tens of cloud examples. Panin et al. [134] presented latent space light probes and proposed the Faster RPNN model, which precomputed some parts of the neural architecture and separated the parts that needed to be inferred at runtime. Zhu et al. [181] proposed a novel approach leveraging a machine learning framework to compress a luminaire's light field into an implicit neural representation. They trained three networks to perform the essential operations for evaluating the complex luminaire at a specific point and view direction, importance sampling a point on the luminaire given a shading location, and blending to determine the transparency of luminaire queries to composite them with other scene elements properly.

While certain high-albedo materials estimate light transport by solving the RTE [178]–successfully applying this to volume rendering–and the theory naturally extends to multi-dimensional problems, these methods still face significant challenges. First, the many-dimensional nature of transmission paths demands an extensive number of samples to reduce estimation variance. Secondly, even with advanced free-path sampling [96] and acceleration data structures [147], constructing a single-path sample remains expensive. In practical applications, MC path tracing can combined with specialized rendering to enhance rendering speed and efficiency.

5.4 NeRF

Neural Radiance Fields (**NeRF**) is a technique designed for view synthesis, enabling the learning of an implicit representation of a static 3D scene from a sequence of 2D images. This representation allows NeRF to render new 2D images from arbitrary viewpoints. At its core, NeRF employs an MLP to model the continuous scene. It takes the 3D coordinates of a spatial point as input along with the viewing direction and produces the color and density information for that point in the scene as output. Its potential applications span diverse fields, including computer graphics, computer vision, and augmented reality, promising extensive opportunities for development and innovation.

Millenhall et al. [115][25] proposed the concept of a NeRF for scene representation. This approach optimized the underlying continuous volumetric scene function using a sparse set of input views, resulting in a remarkable performance in generating new views of intricate scenes. During synthesis, views were generated by querying 5D coordinates along camera rays, with resultant color and density projections onto the image employing classical volume rendering techniques. Their work marked the first effort to propose a continuous neural scene representation capable of presenting high-resolution and realistic views of real objects and scenes captured in natural environments from RGB images.

Li et al. [98] introduced an approach to accelerate NeRF rendering. This method capitalized on information from preceding viewpoints to reduce the number of rendered pixels and ray sampling points along the ray of the remaining pixels. Their pipeline commenced with rendering low-resolution feature maps via volume rendering. Subsequently, a lightweight 2D neural renderer produced images at target resolution, integrating features from preceding and current frames. Gupta et al. [57] presented MCNeRF. This general MC-based rendering algorithm could speed up any NeRF representation, addressing this issue by introducing two key components: (1) MC importance sampling and (2) denoising. At the core of MCNeRF lay the observation that the NeRF volume rendering integration can be evaluated using MC integration over the samples on a ray, with the quality of this approximation was contingent upon the number of samples used (with full raymarching as the upper bound) and the chosen sampling strategy.

NeRF has emerged as a groundbreaking development in computer vision, with over 500 preprints already published on the topic. While this section does not delve deeply into NeRF, readers can refer to surveys [118] and [46] for more comprehensive coverage. NeRF can be trained with a small set of images to generate novel scene views. In contrast, traditional MC path tracing typically demands extensive, high-precision sampling to produce rendered images from different perspectives. There is potential to combine MC path tracing with NeRF's approach, enabling the system to learn variations in lighting and material properties across different viewpoints.

5.5 3DGS

3D Gaussian Splatting (**3DGS**) marks a pivotal advancement in computer graphics and vision. It offers a robust scene representation and innovative view synthesis that operates independently of neural networks, setting it apart from approaches like NeRF. This technique enables real-time, high-quality rendering of radiative fields across diverse scenes and capture styles, all while maintaining training times that rival the fastest methods previously available.

Kerbl et al. [93][26] introduced the 3DGS, focusing on three key innovations that enable exceptional visual quality while maintaining competitive training times. This approach supported high-quality real-time view synthesis (at 30 fps or higher) at 1080P resolution. First, leveraging sparse points generated during camera calibration, the scene was represented using 3D Gaussians. Second, the

[25] https://github.com/bmild/nerf
[26] https://github.com/graphdeco-inria/gaussian-splatting

algorithm incorporated 3D Gaussian interleaving optimization and density control. Lastly, they developed a fast, visibility-aware rendering algorithm that supports anisotropic splatting, which accelerates training and enables real-time rendering.

Malarz et al. [109] introduced a hybrid model called **Visual Direction Distribution** (**VDGS**), which seamlessly combines Gaussian splatting for 3D object shape representation with NeRF-based encoding for color and opacity. This model's Gaussian distribution was characterized by trainable parameters, including position (Gaussian mean), shape (covariance), color, and opacity. These parameters were processed by neural networks, which, in turn, dynamically adjust color and opacity based on both the Gaussian attributes and the observed viewing directions.

3DGS technology is being widely adopted in various fields, including robotics, urban mapping, autonomous navigation, and virtual/augmented reality. For more in-depth discussions and studies, refer to surveys [41] and [108]. Unlike traditional MC path tracing, 3DGS leverages Gaussian distributions to model the volume and surface of objects, simplifying the complexities of volumetric lighting and path tracing. Moreover, the Gaussian distribution effectively approximates light scattering within volumes, making rendering transparent and semi-transparent materials more efficient in path tracing.

5.6 Radiance Caching

The origins of many real-time global illumination technologies can be linked back to the ground-breaking work by Ward et al. [159] in irradiance caching. Modern diffuse interreflections follow this premise, building upon the premise that irradiance transitions smoothly across the scene, using albedo modulation to recover texture detail.

Müller et al. [123] designed a real-time neural radiation caching method for path-traced global illumination. By leveraging a data-driven approach, this method efficiently addressed challenges encountered in caching algorithms, including locating, interpolating, and updating cache points. They achieved the generalization of neural networks through adaptation and opted to train brightness caching during rendering. They employed self-training techniques to minimize the impact of low-noise training targets and simulated infinite-bounce transport by merely iterating few-bounce training updates. Jiang et al. [84][27] created **Deep Radiation Caching** (**DRC**), a radiance caching variant that utilized convolutional autoencoder for global illumination rendering. DRC incorporated denoising neural networks to accommodate a wide range of material types without requiring offline pre-computation or training for each scene, thereby enabling rendering on the CPU.

Indirect lighting is a primary source of significant noise in MC rendering. Radiance caching and a neural network reconstruct and denoise radiance maps. This map can then be interpolated to retrieve the indirect illumination components for the final image. Given the slow-changing nature of indirect illumination, artifacts and bias tend to be minimal, while the overall predicted global illumination remains convincing and noise-free. This approach is a valuable supplement to traditional MC path tracing, enhancing rendering efficiency and quality.

5.7 Photon Mapping

PM is a cornerstone in global illumination techniques, renowned for its robustness and efficacy in simulating caustics and **specular-diffuse-specular** (**SDS**) light paths. These are challenging for unbiased general MC-based methods. Initially, photons are shot from the light sources, bounced in the scene, and stored in photon maps. Subsequently, the radiance of each pixel is computed by estimating the photon density in photon maps.

[27] https://github.com/giuliojiang/pbrt-v3-IILE

Zeng et al. [174] introduced a dedicated learning-based denoiser for **Stochastic Progressive Photon Mapping (SPPM)** [60]. Their core innovation was a dual-network framework that separately denoises caustic components and global illumination components, prior to merging. Critically, a multi-residual block architecture enabled effective handling of multi-scale noise across both low- and high-frequency regions. Choi et al. [27][28] developed a post-reconstruction method for **Progressive Photon Mapping (PPM)**, targeting residual reduction in density estimates. Central to their approach is the fusion of multiple correlated PPM estimates exhibiting multi-level spatial dependencies, utilizing supervised learning-derived weight functions to enforce statistical consistency in the reconstructed solution.

The low-frequency illumination structure of photon maps can be utilized as an auxiliary input to path tracing denoisers, enhancing robustness in global illumination and shadow rendering for dynamic scenes. Alternatively, this structural information can inform the importance sampling strategy within path tracing.

6 Conclusions

In recent years, rapid advancements in deep learning have introduced novel approaches to addressing speed and rendering quality issues in MC path tracing. Traditionally, improving samples efficiency for evaluation (e.g., GPUs, ray tracing hardware, etc.) was the primary focus. However, with the surge in deep learning's popularity and capabilities, new applications have emerged across various domains. In MC rendering specifically, significant research has explored and compared different methodologies, successfully tackling numerous challenges. These studies have uncovered commonalities among methods across different models, highlighting their respective strengths and weaknesses. Such advantages have spurred increased adoption of deep learning for MC rendering tasks, often outperforming traditional physics-based algorithms in performance benchmarks. Concurrently, recent studies demonstrate a symbiotic relationship: advancements in rendering techniques contribute to the evolution of deep learning models, enhancing their quality and efficacy. This reciprocal development cycle promises ongoing improvements and superior outcomes in both fields.

In this survey, we delve into MC path tracing technology based on deep learning. Exploring the rendering equation, we emphasize techniques that optimize sampling, denoise processed images, and enhance rendered quality through upsampling. Additionally, we discuss emerging trends in future development, which, though nascent in research, offer foundational insights and potential for broadening horizons. This opens avenues toward realizing a universally applicable MC rendering in the future. Naturally, the scope of research extends beyond these points, presents a comprehensive overview of the field, remains challenging, and may inadvertently omit certain aspects. Numerous challenges persist within this research domain, yet we anticipate breakthroughs as rendering technology evolves and models advance.

In the predictable future, the demand for MC rendering is still increasing. Realistic visual effects in entertainment offer more profound and authentic artistic expressions across mediums like films, animations, and video games. This technology has enriched experiences in virtual reality, augmented reality, and movie special effects, enhancing various forms of entertainment. In scientific research, particularly in fields like physics and astronomy, MC rendering aids in understanding and predicting complex physical processes, facilitating quicker and more precise observations with potential applications in healthcare. These growing demands propel the advancement of MC rendering toward faster speeds, higher quality rendering, and seamless integration into existing rendering systems. Deep learning-based MC rendering holds promise in achieving these ambitious

[28] https://github.com/CGLab-GIST/consistent-recon

goals, mainly through a unified framework integrating sampling, noise reduction, and upsampling for comprehensive processing.

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